- Vacuum hose

- Power & Distribution
    - 12V Accessory battery (LiFePO4 1-2 Ah)

```
Battery +12V
  ├─10A fuse─> MainRelay(30) ──> ECU + injector + coil + sensors
  └─PumpRelay(30) ──> Fuel Pump +
Speeduino Fuel Pump out ──> grounds PumpRelay(85)
Relay(86) ──> +12V switched
Injector - ──> Speeduino INJ1
Ignition Coil - ──> Speeduino IGN1 / IGBT Driver
```

Total schematic:

```CSharp
                                +----------------+
                                | 12V LiFePO4    |
                                |   Battery      |
                                +--------+-------+
                                         |
                                         | 12 AWG Red (Fused 30A)
                                         v
                                 +----------------+
                                 | Master Cutoff  |
                                 |   Switch       |
                                 +--------+-------+
                                         |
                                         | 12 AWG Red
                                         v
                                +------------------+
                                | Fuse / Relay Box |
                                | 6-way sealed     |
                                +--------+---------+

+-----------------------+-------------------------+-------------------+
                |                         |                         |
                |                         |                         |
        +--------+--------+      +--------+--------+       +--------+--------+
        | Main Relay      |      | Fuel Pump Relay |       | Starter Relay   |
        | (30A SPDT)      |      | (30A SPDT)      |       | (30A SPDT)      |
        +--------+--------+      +--------+--------+       +--------+--------+
                |                          |                         |
                |                          |                         |
                |                          |                         |
```

```
   +-------------+-------------+           16 AWG Red             12 AWG Yellow
   | Speeduino ECU + Sensors   |-------------------------------> Starter Motor +
   |                            |                               (through ESC if
used)
   |                            |
   |    +-------------------+   |
   |    | Fuel Injector(s)  |<--+ 18 AWG Yellow, low-side switching
   |    +-------------------+
   |
   |    +-------------------+
   |    | Ignition Coil      |<-- 16–18 AWG Blue, logic coil or IGBT driver
   |    +-------------------+
   |
   |    +-------------------+
   |    | Crankshaft Sensor |<-- 22 AWG Green/White Shielded
   |    +-------------------+
   |
   |    +-------------------+
   |    | TPS               |<-- 22 AWG Orange
   |    +-------------------+
   |
   |    +-------------------+
   |    | MAP               |<-- 22 AWG Gray
   |    +-------------------+
   |
   |    +-------------------+
   |    | IAT               |<-- 22 AWG Violet
   |    +-------------------+
   |
   |    +-------------------+
   |    | CLT               |<-- 22 AWG Brown
   |    +-------------------+
   |
   |    +-------------------+
   |    | O₂ Wideband       |<-- 16 AWG White (Heater + 0–5V signal 22 AWG)
   |    +-------------------+

        +-------------------+
        | Accessory Relay   |------------------> IMU, GPS, HUD, Camera, Horn
        | 5–10A Fused       |
        +-------------------+
                  |
                  | 18–20 AWG Blue/Green
                  v
```

```
  +--------------+
  | HC-05 + MCU  |--> Tablet RealDash
  +--------------+

+-------------------+
| E-Stop Switch     |
| (cuts master relay)|
+-------------------+
         |
         +--> Cuts power to all fused relays / ECU inputs
```

Electrical Parts List – Nested by Subsystem

1️⃣ Power & Safety

- **Battery**
  - 12V LiFePO4, 1–2Ah
  - Connect to master cutoff: 12 AWG Red
  - Ground to chassis: 12 AWG Black
- **Master Cutoff Switch**
  - Inline with battery +12V
  - Wire: 12 AWG Red
- **E-Stop Switch (NC)**
  - Inline with all critical relay coils
  - Wire: 14–16 AWG Red
- **Fuse / Relay Block (6-way)**

    - Holds fuses & relays
    - Fuses:
        - ECU / Injectors / Ignition: 10A
        - Fuel Pump: 10A
        - Starter: 30A
        - Accessories: 5–10A
    - Wire: same as downstream devices

2️⃣ ECU / Speeduino

- **Speeduino 0.3.7**
    - Power +12V: 20 AWG Red (via Main Relay + ECU power fuse)
    - Ground: 12 AWG Black
- **Arduino Mega 2560**
    - Mounted to Speeduino PCB
- **Inputs / Outputs**
    - TPS: 22 AWG Orange
    - MAP: 22 AWG Gray
    - IAT: 22 AWG Violet
    - CLT: 22 AWG Brown
    - Crank Hall Sensor: 22 AWG Green/White (shielded)
    - O_2: 22 AWG White (+16 AWG Red Heater)

    - Injectors: 18 AWG Yellow
    - Ignition Coil Trigger: 16–18 AWG Blue
    - Fuel Pump Trigger: 22 AWG Black (feeds relay)
    - Starter Trigger (if using MCU control): 22 AWG Blue/Green

---

3️⃣ Fuel System

- **Fuel Pump**
    - +12V via Fuel Pump Relay output 87
    - Fuse: 10A near pump
    - Wire: 16 AWG Red
    - Ground: Battery GND / chassis, 12 AWG Black
- **Fuel Pump Relay**
    - Coil 85 → E-Stop output
    - Coil 86 → Battery +12V
- **Injectors**
    - Low-Side Trigger: Speeduino output, 18 AWG Yellow
    - Power: 20 AWG Red from ECU power fuse
    - Ground: ECU common GND

4️⃣ Ignition System

- **Ignition Coil / CDI**

    - Low-side driver from Speeduino: 16–18 AWG Blue
    - +12V power via Main Relay: 16–18 AWG Red
    - Fuse: Included in Main Relay branch
    - Optional: external small relay if coil >1A
- **Crankshaft Position Sensor**
    - 22 AWG Shielded Green/White
    - Signal → Speeduino input

5️⃣ Starter System

- **Starter Motor**
    - +12V via Starter Relay output 87
    - Fuse: 30A near motor
    - Wire: 12–14 AWG Yellow
    - Ground: Battery / chassis, 12 AWG Black
- **Starter Relay**
    - Coil 85 → E-Stop output
    - Coil 86 → Ignition-on / MCU
- **Starter ESC (if brushless)**
    - Control wires connected to MCU / relay

6️⃣ Accessories / Telemetry

- **Accessory Relay**
  - Coil 85 → +12V via Ignition-on (optional E-Stop)
  - Coil 86 → Ignition-on / MCU trigger
  - Output 87 → Accessories
- **Devices**
  - IMU: 18–20 AWG Red/Black
  - HUD Display / Mini OLED: 18–20 AWG Red/Black
  - GPS Module: 18–20 AWG Red/Black
  - Camera / Horn: 18–20 AWG Red/Black
- **Optional HC-05 Bluetooth Module**
  - Connect to Speeduino serial RX/TX
  - 5V logic from Arduino / Speeduino
  - Wire: 22 AWG

7️⃣ Wiring & Harness Supplies

- **Wire**
  - 12 AWG Red/Black (battery, starter)
  - 14–16 AWG Red (relay coil feeds / E-Stop)
  - 16 AWG Red (fuel pump, ignition power)
  - 16–18 AWG Blue (ignition coil low-side)
  - 18 AWG Yellow (injectors)

    - 18–20 AWG Red/Blue/Green (accessories)
    - 22 AWG Various / Shielded (sensors, triggers)
- **Connectors**
    - Deutsch DT / TE Superseal for critical sensors
    - Crimp terminals for relays/fuses
    - Male/female pins for ECU / sensors
- **Harness Protection**
    - Braided sleeving
    - Heat shrink tubing (3/16–1/4”)
    - Cable ties, grommets, rubber pass-throughs
      \\

PAAAAARTS:

- **12V LiPO4 Battery (w/ BMS and overcurrent protection) **some battery idk**
        -
- Main fuse box & relay box
        - https://www.amazon.com/Malictele-Pre-Wired-Relays-Sealed-Waterproof/dp/B0CYGQKGHW/ref=sxin_17_pa_sp_search_thematic_sspa?content-id=amzn1.sym.9ce0d71e-7aca-4397-8822-32313a4aff8e%3Aamzn1.sym.9ce0d71e-7aca-4397-8822-32313a4aff8e&crid=2H5V5WK1UM8LN&cv_ct_cx=automotive%2Bfuse%2Bbox&keywords=automotive%2Bfuse%2Bbox&pd_rd_i=B0CYGQKGHW&pd_rd_r=0ccad619-9752-4f9c-a397-e825cca8e9e7&pd_rd_w=BWabo&pd_rd_wg=DkI5C&pf_rd_p=9ce0d71e-7aca-4397-8822-32313a4aff8e&pf_rd_r=YBM79NVD1K1HDZCW8GMF&qid=1759884967&sbo=RZvfv%2F%2FHxDF%2BO5021pAnSA%3D%3D&sprefix=automotive%2Bfuse%2Bbo%2Caps%2C204&sr=1-3-6024b2a3-78e4-4fed-8fed-e1613be3bcce-spons&sp_csd=d2lkZ2V0TmFtZT1zcF9zZWFyY2hfdGhlbWF0aWM&th=1
- Main, fuel, ignition, starter relays, +1 (to ECU) **GET FOUR
        - https://www.amazon.com/Bosch-0986332030-Relay/dp/B00BHIM4N0/ref=sr_1_3?crid=BGLPG6GF9N7Q&dib=eyJ2IjoiMSJ9.XPEqjgc_SaogMzDJ

_6pjIJOkdotmROLds-fhNYCFSRpgm6-MZwHdaVL2wg179iRALKoRrWO2HftzRYREI59EydOU4_71FkI-wiJcofJA8ib_xxloiDfJ862q5McWQ7dyYA8OTeCyC95pZgDFxWLzUAT6Cxd6cwxIiP1LeDjGHjQC4sMbdApo3A3hWMGDXL62uquE3oo7url6X_NlMNqz4Ot3UU6Mht_XKwhtkmX5wQl2eM-pn5Ehns1ViQCJ85cdCNgMeulPYJamHvQMwclC2waPOJwXQjjOcV_L5oCnCSw.2R0y7E6n32U7Tc2AK_UGtvRN4aG31h4pdcQzQ6KqyMM&dib_tag=se&keywords=4+pin+bosch+relays+40A&qid=1759884191&s=industrial&sprefix=4+pin+bosch+relays+40a%2Cindustrial%2C145&sr=1-3

- E-Stop (connect to signal line for main relay)
    - https://www.amazon.com/API-ELE-Warranty-Emergency-Latching-Stianless/dp/B08QD31NNC/ref=sr_1_2_sspa?crid=3G5FFCEWREA91&dib=eyJ2IjoiMSJ9.o0JLMga4CanfnAaJZnGvPmo8cMM38jChPtEgBaRhodWKAtfYG2sZ6NugYS5MTmdwmcF9rJUZ1Ok1gAzY-Gpa-ApzU-QPTctDBXxXnb8nJzA42jE24lzyD5k_zYC1TzuRRT1DRGbI5iLHiVpuzJpXf6g0-VxnOMiU_WRCoFMp9tI14V_8IU04TLUWQf4T89V1UVirAMYJH7nSDJnaIy-UNLZAgkiZs-lH3T5TADT--js.zilyJC3X59NI8hIOy3OikqIApiORAC9azPxZBqZS7T8&dib_tag=se&keywords=emergency%2Bstop%2Bpush%2Bbutton%2Bautomotive&qid=1759883196&sprefix=emergency%2Bstop%2Bpush%2Bbutton%2Bautomotiv%2Caps%2C170&sr=8-2-spons&sp_csd=d2lkZ2V0TmFtZT1zcF9hdGY&th=1
- 12V-5V convertor (for tablet) [3A fuse]
    - https://www.amazon.com/Converter-Module-Output-Adapter-Regulator/dp/B08RBWX2GL/ref=sr_1_2_sspa?crid=980WU7L0VRAI&dib=eyJ2IjoiMSJ9.HxKTW0eyVH9xdmECVZ0jwP1hnqZ0YLrX33tLWmshWiAj430sqsfUZKdYqSj6XqW9KttgEpGZwl6OSc5wZxtVDTW9besYZTQNYZdCzhF9ErowIhZ3AWHXjZEtzb_YlqXz7RyWOU9eP0Gu0dDmVYnZAv1Nxin7eTJ4svPEkIBhOPA6sCHp27s1cAyt1O5yGkgbet5HCXCUdVHO5QvAGlIQAu-sIR6gv9uUkIhV5PR3a74.WjvNA9GEC6PXa5oVNugpibUZOIEtFGED7wYVzY9-9YY&dib_tag=se&keywords=5v%2B3a%2Busb%2Bbuck%2Bconverter&qid=1759883591&sprefix=5v%2B3a%2Busbc%2Bbuck%2Bconverter%2Caps%2C545&sr=8-2-spons&sp_csd=d2lkZ2V0TmFtZT1zcF9hdGY&th=1
- Fuse assortment (5A-30A)
    - https://www.amazon.com/Liamner-Assortment-Automotive-Motorcycle-Accessories/dp/B0DSJQTBVY/ref=sr_1_1?crid=2UAD0T694R14E&dib=eyJ2IjoiMSJ9.oO_4cubOHzKcIba3OUDPdBq5itrKdVwAaweh8G8WdyXY_k3peOn2UvqdKm4-KeyhsO93sE1x2NUi4XvdQBKWENYU8HDVRTM-iM7DZoVAi4YQ7BH2KIIXGBoLJdYRIQCJuNfv0LA2f6L-7b5UPXH8tgi-0Kw6FCxEqmEKSL5qeD7cG4XH-ok7DU_j4lqB_mGgoPiuIualk6MRItoIkkT01HMiIHq171FdEhqpAI7pq0c.TqBq9xa4UnOCYl8BaOFEHiLZZw6a00PaM4aPH3KboTQ&dib_tag=se&keywords=AOC+mini+fuse+assortment&qid=1759887110&sprefix=aoc+mini+fuse+assortment%2Caps%2C167&sr=8-1
- **Fuse for starter motor?**
    -
- MAXI fuse and holder (60A for battery)

        - https://www.amazon.com/Fastronix-High-Current-Covered-Holder/dp/B071CWH241/ref=sr_1_1_sspa?crid=2PN059TTE44XL&dib=eyJ2IjoiMSJ9.j6OyDbObYc7h0ohrCkImUPGt8pdZfLrDFRDCbYXeHqOmv13DuO7B0tmnOaXvVvYOAWoR03BNsoLqlYHP-z82-hqLmBUi4kxpRsbyHbDuOqf0W0F5ahnzg1IufPe0_q6yWC92_R5snAh6xwt-0-jTUQ1G9GMbdngbx_E_WDgK8L-O-7AukdFz7v8afGK4i7gaWwAkEc-KzsRWf4FpOts1__GrNov5VMJy4FBCi9ZuXv3m2TBPcqU-yCG2Tfqz_Eq5xy5Bccz53BqsCMLU0dZV2P-oHS7qQ_w4CZSv_wCPfvw.s3P_oNkJUBQzijacT2YlqlStRH0rLb-tMhsJGJ6tWZM&dib_tag=se&keywords=4+gauge+maxi+fuse+holder&qid=1759885819&s=industrial&sprefix=4+guage+maxi+fuse+hold%2Cindustrial%2C149&sr=1-1-spons&sp_csd=d2lkZ2V0TmFtZT1zcF9hdGY&psc=1
- Ground bus bar (gnd for all the big boy current drinkers ex: battery, main relay, starter, fuel pump, ignition coil, etc)
        - https://www.amazon.com/Blue-Sea-Systems-Common-10-Gang/dp/B000K2MA9C/ref=sr_1_1?dib=eyJ2IjoiMSJ9.T-8-2hx7zz9L5WV-pz77KWrQcZfKBWz_cEybNXFxLlBiOF9ItJszcCHI9J-UgEDl0dgTPZWOOte7Qj9zO_-AXn81TKoiT-XlDNxW9a483duv2sCJ0uhTRZAMA8yJd_6d5NJktF2OSnb7veSAa91tsbDvXc8Khnmh-oKH0PjKTF5IpMUr_wNpK1_BjzYcHXJo9sea5TyGd9FWTFyROoUiB2JJM3rmfct5Z9jAt8608B1yf4tKwMNtOx1bf9goUAQSVTXeVB-E71EPEZy8t9An5igc07s8GqcQmO0IYiqjo50.sg8zklWyqZG9ACXOWXQYIM4hfMfPpOX3ROyA3xjvVBg&dib_tag=se&hvadid=2419124036 94&hvdev=c&hvexpln=0&hvlocphy=9032030&hvnetw=g&hvocijid=9473815932346308800--&hvqmt=e&hvrand=9473815932346308800&hvtargid=kwd-6778157821&hydadcr=24662_10400876&keywords=blue+sea+2301&mcid=44dfa469103b3366b2c2f3116110850d&qid=1759886674&sr=8-1
- Terminal Block
    - (to split ECU power to all sensors and HC-05 and ignition relays)
        - https://www.amazon.com/Joinfworld-Terminal-Position-Pre-Insulated-Jumper/dp/B0BN55WVDB/ref=sxin_16_pa_sp_search_thematic_sspa?content-id=amzn1.sym.9ce0d71e-7aca-4397-8822-32313a4aff8e%3Aamzn1.sym.9ce0d71e-7aca-4397-8822-32313a4aff8e&cv_ct_cx=insulated%2Bterminal%2Bblock&hvadid=241904436466&hvdev=c&hvexpln=0&hvlocphy=9032030&hvnetw=g&hvocijid=12081548633357188257--&hvqmt=e&hvrand=12081548633357188257&hvtargid=kwd-1194057008&hydadcr=24630_10399594&keywords=insulated%2Bterminal%2Bblock&mcid=66f90a89f488390f8d4f5ba1b62b9963&pd_rd_i=B0BN55WVDB&pd_rd_r=66624062-985e-4507-91fc-76a4ed1845a3&pd_rd_w=9mHBV&pd_rd_wg=xR0ng&pf_rd_p=9ce0d71e-7aca-4397-8822-32313a4aff8e&pf_rd_r=CMT6YTPYSGJ2YR5PZ857&qid=1759886961&sbo=9ZOMT9Jm0JH%2Ft%2BWi68iDSA%3D%3D&sr=1-2-6024b2a3-78e4-4fed-8fed-e1613be3bcce-spons&sp_csd=d2lkZ2V0TmFtZT1zcF9zZWFyY2hfdGhlbWF0aWM&th=1
    - Common gnd bus (for small sensors)
        - https://www.amazon.com/Bonsicoky-Terminal-Positions-Electrical-Grounding/dp/B097R5J66D/ref=sr_1_7?crid=16RS9BLMV1BPN&dib=eyJ2IjoiMS

J9.o3aAIQFhu_-oQy_cjH_gGr-I2ZsUD2van3ZxTJFIcuCGtwjsV_0a02c06xmGgAsweLC_Z7s8BrkThGN9AE2FHt6aSQ5IIX0xm6AywIxyYmkjc037Wnkoc7aHFneKdHEK9K7xrVlhkaOnULBed569N-pp-0vxnDiirp3qnsHDPBmeOkuzUH1762MrlZQMN0zGQIZrdOFa-srUtJsQnGndByVy5Y_iN0x7BqRJwkpjwacSbjMiRnEtuu5OaBVW9xk3zcHL_uoTfN8-jhZxIzkypbC-lzw2f7pjCWYmur6nO-Q.0BQxbNhj80H8sG3uCdPtALvmOI-11KQYgsoqWwOeHoE&dib_tag=se&keywords=ground+bus+bar&qid=1759886729&sprefix=ground+bus+ba%2Caps%2C175&sr=8-7

- Weatherproof connectors (2,3,4 pin) **NOT for starter, battery, ignition coil, fuel pump
  - https://www.amazon.com/Amliber-Ratcheting-Connectors-Automotive-Electrical/dp/B0BZBZHVQ6/ref=sr_1_1_sspa?dib=eyJ2IjoiMSJ9.1ggx0u9O8ELorYgTbvtMU2czdcwaFc6AGmgtEHUKTuP9BaAg0ne6URIOA9saY5EdVU5WIdOAgs_0TiXBsF1LgMzY8n-skX_APQJXcLPIy8kdjoSHtn_zRdM9TY-rmljTZVdQhqaLR1CDuGMzAnfMxUcfN6BGmdwt5Fa-O5jgd5LWTOkXMDiTed4VdjJUsZi6d2bUyfEGK8lH8jY5wxCZjeV2T30tmA5C5OToGBfZSgM.FDl0ZpkyE-u2IbWTmRl7XDYJEIyOgQPVhL9bX9JeItI&dib_tag=se&hvadid=703719639063&hvdev=c&hvexpln=0&hvlocphy=9032030&hvnetw=g&hvocijid=16756300949084176163--&hvqmt=e&hvrand=16756300949084176163&hvtargid=kwd-377730786898&hydadcr=11111_13663930&keywords=deutsch+style+connectors&mcid=107fbdab59613c62889eed84540b970b&qid=1759887500&sr=8-1-spons&sp_csd=d2lkZ2V0TmFtZT1zcF9hdGY&psc=1
- **Ring Terminals**
  - **8 AWG for battery**
- Butt splice connectors (heat shrink typed)
  - In case a wire is too short or we have some cuts (use solder+silicone for high current loads the big 4)
    - https://www.amazon.com/TICONN-120PCS-Heat-Shrink-Connectors/dp/B0CT3YCD5B/ref=sr_1_2_sspa?crid=2Y4W1LODQ784C&dib=eyJ2IjoiMSJ9.h2nMOlQHGbm4s-HtC2fI2Mn6QzWhqcSuMnBS7vmVtfj7reFlekePrjOnEPE0T99HzLrJJNGTbhKJvOm_-Ez1LZ69GXCr6ofYBboRI_fx7Wp9PXsTXcyhKwLx7i14xK2z1-PewJSGXrPHWb5LsKe_kELwcM-1Z9T7aCsYowmQQaaOwdLT8mZtj0QZKZi5zMUg6ar-F0_TrM_N-qUhVorWlDfxydkMC7BDQEWOQjdQR1s.eirYoBGIm_v91z-uYityp1qdKbkgUVmnK7849dazY90&dib_tag=se&keywords=butt%2Bsplice%2Bconnectors&qid=1759888070&sprefix=butt%2Bsplice%2Bcpnnector%2Caps%2C181&sr=8-2-spons&sp_csd=d2lkZ2V0TmFtZT1zcF9hdGY&th=1
- **Wires**
  - **16 AWG**
  - **14 AWG**
  - **12 AWG**
  - **8 AWG**
- Wire loom
  - https://www.amazon.com/Keco-100ft-Expandable-Braided-Sleeving/dp/B07K1XJNJN/ref=sr_1_12_sspa?crid=2OZ7V2MNLQ6T3&dib=eyJ2IjoiMSJ9.HF9US2Rw9

_GcjXVTBsnnylxO8rXuyvDd9CJXLN1dBDwPHXRQ134gVffN5JSnXL5Ig-Bp0lovlCQUDsoAUzaELJ8_rCftOajxk95Zv5_x7ri-YDqtbYowp2Khj6_by7HNDUTme49WMpIX9sorwrS_ahVBW5CEr9_1eAIaKI4gRMUkxxGbPDXJWQOjO2dwwBTr3qJ6T_kZd8ytMnHG1c_snYg0bb9tHprbSWjys5pHL0w.XJ5h0lHsc4v_z16W0fzsh5b5Txc_IgPtqekMdtYSJlw&dib_tag=se&keywords=techflex%2Bloom%2Bautomotive&qid=1759888848&sprefix=techflex%2Bloom%2Bautomotiv%2Caps%2C149&sr=8-12-spons&sp_csd=d2lkZ2V0TmFtZT1zcF9tdGY&th=1

- Heat shrink tubing assortment
  - https://www.amazon.com/108PCS-Heat-Shrink-Tubing-Kit/dp/B0FB8PD9NF/ref=sr_1_4?crid=16UMJWRLLPLQB&dib=eyJ2IjoiMSJ9.Spf3dMvbC7akN-kdMAHYXBJMzzPLI7-4xPIicm6fzcjgFXFvESscd91t4zK26lEJTAE_Ive82LmqkIEQqNxadEe8kts4JxAU0Pb9c33_jOddnzNFUP0lRF9KNzMCU9dDsfXsrJ4Hmy6tmDR94TJFCcvb4o-AF97wonaRw3itgwd9cu_AwmPU8faqfBw3kerOaiqMJAGdfJkCiQkvd3MNsl77dMhn6QpNs5sBhJv7Ajw.aEplN0LRMrwJIMhf3a2GQ84KmMHfjLaUV3J_0uPxlbo&dib_tag=se&keywords=color%2Bheat%2Bshrink%2Bassortment&qid=1759889009&sprefix=color%2Bheat%2Bshrink%2Bassortment%2Caps%2C150&sr=8-4&th=1
- Neutral cure silicone
    - https://www.amazon.com/BTAS-Dielectric-Automotive-Electrical-Connectors/dp/B0D6R543V2/ref=sr_1_1_sspa?crid=193SP22JTJRBH&dib=eyJ2IjoiMSJ9.Z80JQzS5p9_BJfAW5zODdKGRWqBLUcrSsEvCrW-q6So9UjJB8BPjIz64AfkIdUQMg1axJOfK63N2REXhhN1BL_MU6oZFF47cDdJysieDkviKeyAOrtvaDxkNA3cSqk-0hemI5krS4vBrAfuoY3ruGEsbQImrwVigZN42uQZ5lVL3gBm0r9X2TKh1yfS6WzDFam9fzm-dMgCUTU-LJMCiOkPOpwpuUNLwqGut8PCif7Y.4751hMK7ELW831SXGCY_57PYj5BGMBwtKBNxn2heMv0&dib_tag=se&keywords=electronic%2Bsilicone&qid=1759889237&sprefix=electronic%2Bsilico%2Caps%2C166&sr=8-1-spons&sp_csd=d2lkZ2V0TmFtZT1zcF9hdGY&th=1
- Fireproofing silicone
  - https://www.amazon.com/Visbella-Temperature-Waterproof-Combustion-Automotive/dp/B0CPNZNQC4/ref=sr_1_2_sspa?crid=NSCK4A5N42TH&dib=eyJ2IjoiMSJ9.CC41cRqb5Un1Oyc3A4BxudsGjV14kJ2j0IlZ-QyAqBwTkDnj510G-Nu94b9VYAL-65Hhs8DwRu1LSFKiIYXEKkzZLr-LPeA88z79s-8731N2UK7gdo7UHihJF1SJwIGNAPaBwyDUgEu7fMkuEybEvHuDLRZyItReXrXu88vhOSuHyXE0eMQnZ-ch6Pvgym0iLmTVsmTV0ebciaqPMKEibhP4O3fJAkj8_3YRsWwOADM.9eP6iTOxUfpKh7DYh-XDV2bOLUVV2U6EuO4jYPNgxV4&dib_tag=se&keywords=RTV%2Bhigh%2Btemp%2Bsilicone&qid=1759889310&sprefix=rtv%2Bhigh%2Btemp%2Bsilicon%2Caps%2C166&sr=8-2-spons&sp_csd=d2lkZ2V0TmFtZT1zcF9hdGY&th=1
- Wire grommets (to pass through chassis)
  - https://www.amazon.com/Double-Sided-Rubber-Grommets-Assorted-Waterproof/dp/B0B5VYYSCM/ref=sr_1_4?crid=2N0FC9A2VKCGF&dib=eyJ2IjoiMSJ9.3SOUVw7CBrC8PiZUE21WfaUU51w9B_UBACTKcqdjygMYzYBXDQtsopCvHtXHrMH9BFmg1slKm_m9rqXcd6nx2RU00N8mIhq6Z4az-hj7g1rFKQpzyXcMoT4WxUuCwxxGOWEZAQZkc6mruKeFpLs6y1Kr1aOI1QXf74qrZemUR11gmWpaVdoII-G7L

B7rGyG5PXoaCJYmlTM4Voc6WOhWRF_mXBU_amE-V-uKh_PCWxk.-TKhooNmu9NkNOOkLtq5xzSXLe3WUY8PGt4W3qK5SIw&dib_tag=se&keywords=wire%2Bgrommets&qid=1759889343&sprefix=wire%2Bgrommet%2Caps%2C174&sr=8-4&th=1

- Main battery cutoff switch
  - https://www.amazon.com/Current-Battery-Disconnect-Isolator-Vehicle/dp/B0B7DR5H4H/ref=sr_1_2_sspa?crid=3NMNTZPJEN1CM&dib=eyJ2IjoiMSJ9.GouYDG6NBYkcrXCmGRTFWFW_1fj0trvH4JF1b1ojwaA0k9gAfwQa4dKn4ICq0BkViPE82bRzwFpcayiafmJ37nBtPObK1gW8cGitW1WrjB6gtTn2OPOz_s5Ias4R0FMg0Z3XCVZPRq_JPed1BfBuBV4OHNyu5pj7xsUZxrF5SWxRCTGyJm-ETROlKbDfFZOGRwK336itQgbzR0keuOdReoqE8D_Dinu-Kz_87ZF013Y.k-_qxZe28S5i-bZT0qkUCEavM1MGbCwksUQz3gxEJ2Y&dib_tag=se&keywords=300A%2Bbattery%2Bdisconnect%2Bswitch&qid=1759890339&sprefix=300a%2Bbattery%2Bdisconnect%2Bswitch%2Caps%2C183&sr=8-2-spons&sp_csd=d2lkZ2V0TmFtZT1zcF9hdGY&th=1

Extra:
~~Crimping tool~~
Multimeter
ECU enclosure box (are we making this?)

Final Wire List + **8 AWG wire for battery and starter

AWG	Color	Total Approx Length	Use
12 AWG	Red	8 ft	Battery + main fuse + main relay + starter
12 AWG	Black	4–5 ft	Battery negative → main ground bus / high-current grounds
14 AWG	Red	6 ft	Main relay → fuel pump + ignition coil
14 AWG	Black	3 ft	Medium-current grounds (fuel pump, ignition coil if not chassis grounded)
16 AWG	Red	6–7 ft	ECU 12 V power input + small accessories + 5 V step-down for tablet
16 AWG	Black	12 ft	ECU / sensor grounds (split to sensors and relay coils)
16 AWG	Blue	15–20 ft	All low-current signals: TPS, MAP, IAT, CLT, Hall sensors, HC-05, relay coils

⚙️ Core Power & Distribution

Item	Notes / Verdict
✅ **12V $LiFePO_4$ Battery**	Perfect. 1.5–3 Ah minimum for EFI. Get one with **BMS and overcurrent protection**. Example: Antigravity or Shorai.
✅ **Main Fuse Box**	Essential. 6–8 way ATC/ATO fuse block with **positive bus** (Blue Sea Systems or Littelfuse).
✅ **Main Relay (ECU)**	Good. 30–40 A automotive relay, standard SPST (Bosch-style). Triggers from ignition switch or E-stop return.
✅ **Fuel Pump Relay**	Yes — same type as main relay (use socket base with pigtail).
✅ **Ignition Coil Relay**	Optional — if Speeduino coil output drives an igniter directly, relay not needed; but keep it for safe power control.
✅ **Starter Relay**	Mandatory — use a **heavy-duty solenoid relay (40–100 A)** depending on your starter motor draw.
✅ **E-Stop**	Great — choose **normally closed (NC)** style; wire inline with main relay trigger.
✅ **12 V → 5 V Converter (Tablet)**	Good. Choose **buck converter (5 A rated)**, 15–20 W output. Example: DROK 12V→5V 5A USB module.
✅ **Fuse assortment (5A–30A)**	Yes — standard ATO or mini-blade fuses.
✅ **Ground Bus Bar**	Yes. Use for all return grounds to battery/chassis. Get one with 6–10 screw terminals.
⚠️ **Terminal Block**	Keep **one** only. Use to split **+12 V switched power** for sensors, HC-05, and ignition enable. Make sure it's rated for 15 A+ and insulated.

🔌 Connections

Item	Notes / Verdict

✅ **Weatherproof Connectors (2,3,4 pin)**	Excellent. Get **Deutsch DT** or **Superseal 1.5** kits.
2-pin → IAT/CLT	Perfect.
3-pin → TPS, MAP, HC-05	Yes.
4-pin → Injector/Coil	Yes — or 2-pin if each gets its own plug.
✅ **Ring Terminals (12 AWG)**	Keep — for battery, ground bar, and heavy gauge wires only.
✅ **Butt Splice Connectors (heat-shrink type)**	Absolutely. Get **marine-grade**.
✅ **Inline Fuse Holder (for ECU Power)**	Yes — 5A fuse inline before ECU's +12V input.

🧵 Wiring

Gauge	Purpose	Notes
12 AWG	Battery → Main Relay, Starter Relay	Silicone-insulated, high current.
14 AWG	Relay → Fuse Box feeds, Fuel Pump	Handles 10–15 A safely.
16 AWG	Ignition Coil, Injector Power	Perfect balance of flexibility & capacity.
18 AWG	Sensors, ECU signal lines	Standard automotive control wire.
✅ **Wire Loom**	Yes — PET braided or split corrugated for protection.	
✅ **Cable Labels**	Highly recommended — saves debugging headaches. Get adhesive heat-shrink ones or write-on tags.	

🔧 Tools / Accessories

Item	Notes / Verdict

✅ **Crimping Tool**	Must be a **ratcheting crimper** compatible with Deutsch/insulated terminals. IWISS and Wirefy make good ones.
✅ **Multimeter**	For continuity checks, ground verification, voltage testing.
✅ **ECU Enclosure Box**	Absolutely — either a **3D-printed or aluminum case**. If mounting near engine, use a sealed box with grommets.

⚠️ Add These (You'll Need Them)

Item	Purpose
Heat-shrink tubing assortment	For sealing soldered joints and strain relief.
Neutral-cure silicone (sensor safe)	For sealing joints and connectors.
Wire grommets / strain reliefs	Where the harness passes through panels or boxes.
Relay sockets with pigtails	Cleaner install for 4-5 pin Bosch relays.
Grounding ring / chassis lug	To tie the ground bus to frame solidly.
Main battery cutoff switch	For absolute power isolation when servicing.